

R^n
(Co-
or-
di-
nate
vec-
tors)

$n \times$
 1
 E^n
(Eu-
clidean
vec-
tors)

M^n
(S-
pa-
tial
mo-
tion
vec-
tors)

M^6
 F^n
(S-
pa-
tial
force
vec-
tors)

F^6
 $m \times$
 $nm \times$
 $n \mathbf{R}^{m \times n}$

$\mathcal{V}$

V

V^*

V^*

V

V^*

$\mathcal{U}$

$u \in$
 $V^*, v \in$

V

M^n

F^n

$m \in$

M^6

$f \in$

F^6

$\mathcal{D}$

$\{d_1, \dots, d_n\}$

$\mathcal{U}$

$\mathcal{E}$

$\{e_1, \dots, e_n\}$

V

$d_i \cdot e_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$

(1)

$u \cdot v = \underline{u}^T \underline{v}$

(2)

$\underline{u}$

$\underline{v}$

$\mathbf{X}$

$\mathbf{X}^*$

$\mathbf{X}^* = \mathbf{X}^{-T}$

(3)

V

n

$M^n V^*$

F^n

$\langle \cdot, \cdot \rangle :$

$V^* \times$

$V \rightarrow$

R

Bilinearity

$\forall a, b \in$

R

$\langle au_1 +$

$bu_2, v \rangle =$

$a \langle u_1, v \rangle +$

$b \langle u_2, v \rangle$

Non-

degeneracy

$u \in$

V^*

$\langle u, v \rangle =$

0

$=$

$\mathbf{0}$

V

V^*

$\mathcal{E}$

$\{e_1, \dots, e_n\}$

V

V^*

$\mathcal{D}$

$\{d_1, \dots, d_n\}$